2.13 Passive Transport

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What you will learn

- How molecules can pass through a cell membrane in passive transport, active transport, and vesicle transport
- What happens in **diffusion**, osmosis, and **facilitated diffusion**

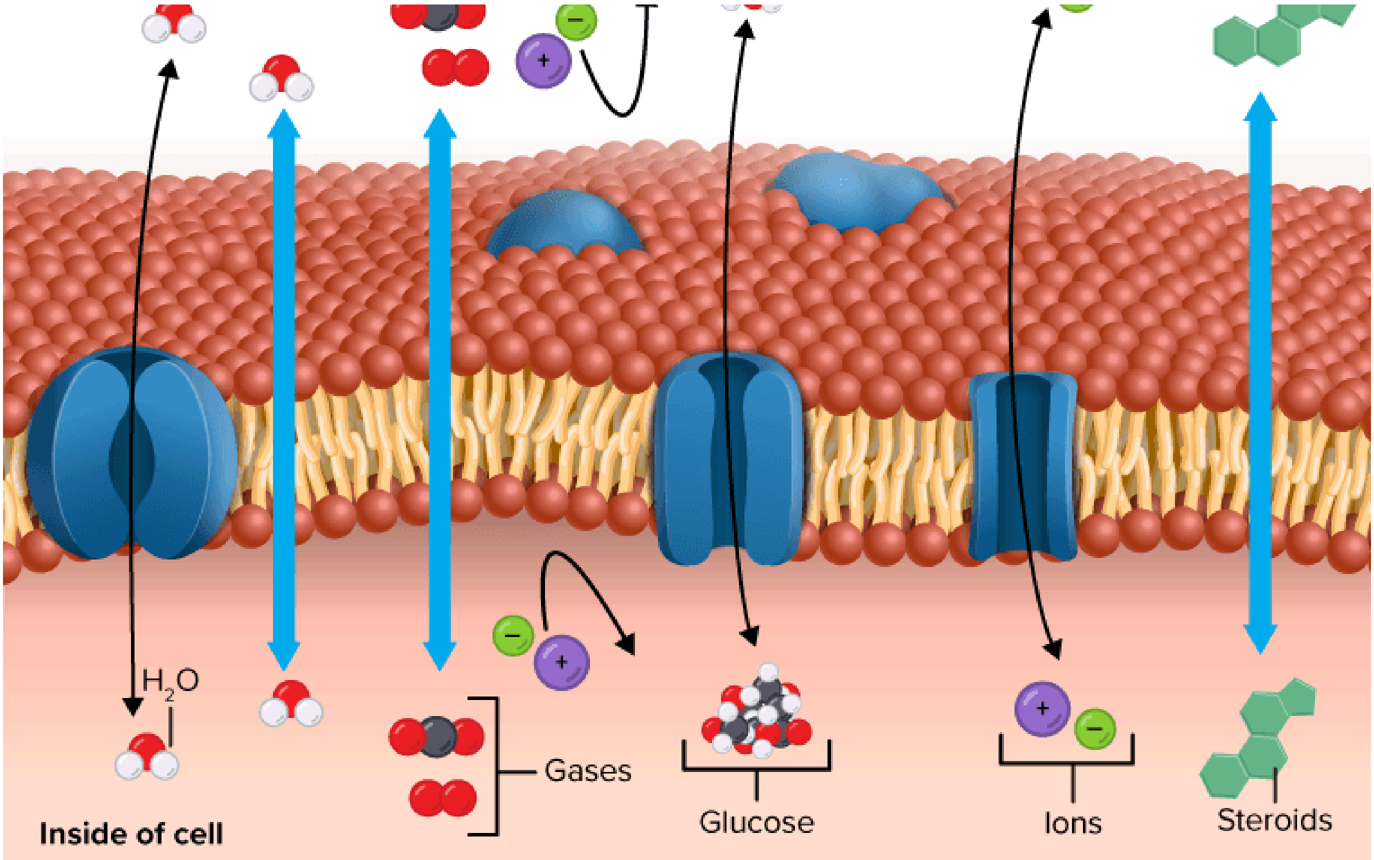
[Figure 1]

What will eventually happen to these dyes?

They will all blend together. The dyes will move through the water until an even distribution is achieved. The process of moving from areas of high amounts to areas of low amounts is called diffusion a type of passive transport.

Passive Transport

Probably the most important feature of a cell's phospholipid membranes is that they are **selectively permeable** or **semipermeable**. A membrane that is selectively permeable has control over what molecules or ions can enter or leave the cell, as shown in the **Figure** below. The permeability of a membrane is dependent on the organization and characteristics of the membrane lipids and proteins. In this way, cell membranes help maintain a state of homeostasis within cells (and tissues, organs, and organ systems) so that an organism can stay alive and healthy.



[Figure 2]

A selectively permeable membrane allows certain molecules through, but not others.

Transport Across Membranes

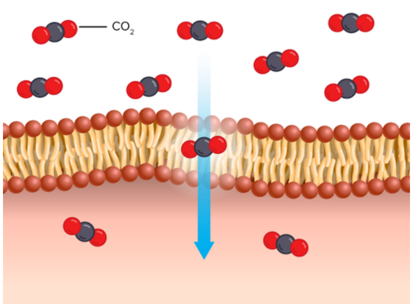
The molecular make-up of the phospholipid bilayer limits the types of molecules that can pass through it. For example, **hydrophobic** (water-hating) molecules, such as carbon dioxide (CO_2) and oxygen (O_2), can easily pass through the lipid bilayer, but ions such as calcium (Ca^{2+}) and polar molecules cannot. The hydrophobic interior of the phospholipid bilayer does not allow ions or polar molecules through because these molecules are **hydrophilic**, or water loving. In addition, large molecules such as sugars and proteins are too big to pass through the bilayer. **Transport proteins** within the membrane allow these molecules to pass through the membrane, and into or out of the cell. This way, polar molecules avoid contact with the nonpolar interior of the membrane, and large molecules are moved through large pores.

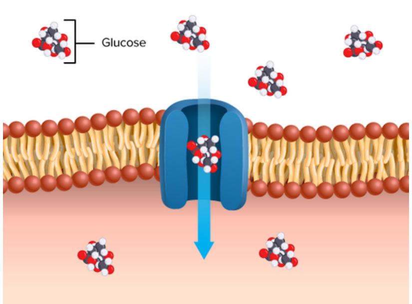
Every cell is contained within a membrane punctuated with transport proteins that act as channels or pumps to let in or force out certain molecules. The purpose of the transport proteins is to protect the cell's internal environment and to keep its balance of salts, nutrients, and proteins within a range that keeps the cell and the organism alive.

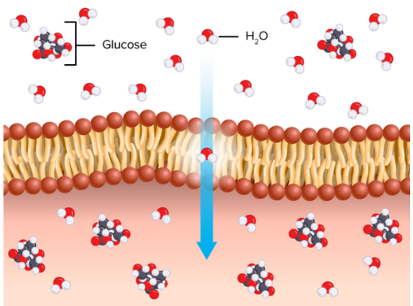
requires that the cell uses energy to pull in or pump out certain molecules and ions and is called **active transport**. The third way is through vesicle transport, in which large molecules are moved across the membrane in bubble-like sacks that are made from pieces of the membrane.

Cell Transport
Amoeba Sisters

Watch on

- 

[Figure 3]
Diffusion
- 

[Figure 4]
Facilitated diffusion
- 

[Figure 5]
Osmosis

Passive transport is a way that small molecules or ions move across the cell membrane without input of energy by the cell. The three main kinds of passive transport are diffusion, osmosis, and facilitated diffusion (see the **Figure** above).

Diffusion is the movement of molecules from an area of high concentration of molecules to an area with a lower concentration. The difference in the concentrations of

- **Facilitated diffusion** is the diffusion of solutes through transport proteins in the cell membrane. Even though facilitated diffusion involves transport proteins, it is still passive transport because the solute is moving down the concentration gradient.
- **Osmosis** is the diffusion of water molecules across a selectively permeable membrane from an area of higher water concentration to an area of lower concentration.

DID YOU KNOW?

Dialysis involves the diffusion of solutes from blood across a semipermeable membrane. Dialysis is a procedure to remove waste products and excess fluid from the blood when the kidneys stop working properly. It often involves diverting blood to a machine to be cleaned.

Summary

- The cell membrane is selectively permeable, allowing only certain substances to pass through.
- Passive transport is a way that small molecules or ions move across the cell membrane without input of energy by the cell.
- The three main kinds of passive transport are diffusion, osmosis, and facilitated diffusion.

Review

1. Define semipermeable.
2. What is diffusion?
3. What is a concentration gradient?
4. What is meant by passive transport?